ANALOGUE ELECTRONICS

COMPONENTS

- Battery
- Resistor
- Switch (SPST)
- Lamp
- Diode
- Motor
- Variable Resistor
- Light Dependant Resistor (LDR) (L.U.R.D.)
- Thermistor (T.U.R.D.)
- Light Emitting Diode (LED)
- Buzzer
- (A) Ammeter
- (V) Voltmeter

* = Input Transducer

* = Output Transducer

CALCULATIONS

Ohm's law: $V = IR$

RESISTANCE

Series: $R_T = R_1 + R_2 + R_3 \ldots$

Parallel: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$

R_2 resistors in parallel $= \frac{R_1 \times R_2}{R_1 + R_2}$

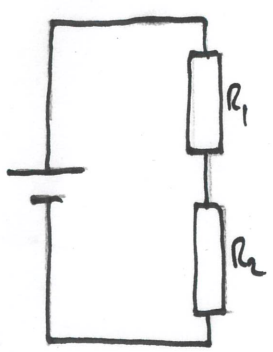
Electrical Energy, $E_E = VIt$

Electical Power, $P = VI$

SEE DATA BOOKLET FOR FORMULAE

TYPES OF CIRCUITS

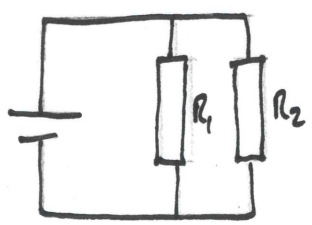
Series



I_T = same at all locations

$V_T = V_1 + V_2 + \ldots$

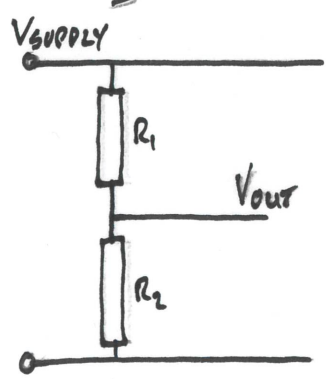
Parallel



$I_T = I_1 + I_2 + \ldots$

V = same in every branch

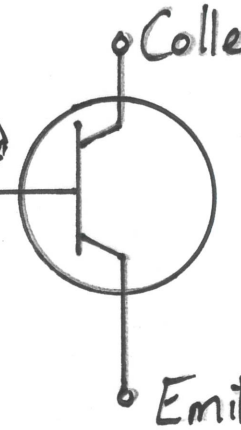
Voltage Divider



$V_{OUT} = \frac{R_2}{R_1 + R_2} \times V_{SUPPLY}$

$\frac{V_1}{V_2} = \frac{R_1}{R_2}$

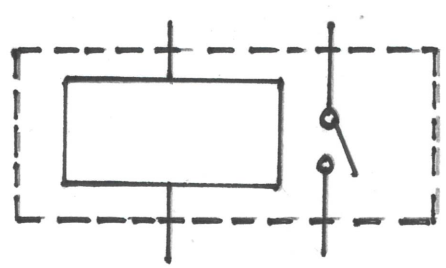
TRANSISTORS



(NPN Type)

Turns ON if ~0.7V applied to base.

RELAYS



DIGITAL SWITCHES